

Karthik Reddy Duddukunta

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Professional Summary

Data Science graduate student specializing in ML, LLM/RAG systems, and computer vision, with hands-on experience building end-to-end pipelines from data ingestion through deployment. Track record of applying predictive modeling, causal inference, and geospatial/CV analytics to institutional-scale datasets, with peer-reviewed publications and technical leadership experience.

Education

Master of Science in Data Science 2025 – Present

Montclair State University, New Jersey, USA

CGPA: 4.00 / 4.00

Bachelor of Technology in Computer Science and Engineering 2020 – 2024

SRM Institute of Science and Technology, Chennai

GPA: 9.56 / 10

Skills

Technical: Python, Exploratory Data Analysis, Machine Learning, Deep Learning, NLP, LLMs, RAG, SQL, Spatial Statistics

Tools: Power BI, QGIS, ArcGIS, Docker

Frameworks: TensorFlow, Scikit-learn, Spacy

Work Experience

Research Associate—Clean Energy and Sustainability Analytics Center

Montclair State University, New Jersey, USA

May 2025-Present

- Developed predictive analytics models using Cox Proportional Hazards (Survival Analysis) on 5000+ employee records to forecast milestones and retirement eligibility, improving workforce planning accuracy.
- Built end-to-end data pipelines spanning ingestion, preprocessing, feature engineering, model training, and validation on large-scale institutional datasets.
- Performed EDA and applied ML/statistical modeling to accounting and sustainability analytics, identifying climate-driven energy consumption trends to support institutional reporting and planning.
- Mentored 5+ high school and master's students and led five master's-level project teams, providing technical leadership in model selection and evaluation.

ElectricAI — Energy Cost Intelligence & Predictive Analytics Platform May 2026 – Present

Python, FastAPI, React, TypeScript, LangGraph, LangChain, PostgreSQL, Redis, Celery, Docker, XGBoost, Prophet, SARIMAX, DoWhy, EconML

- Independently designed and developed a full-stack energy intelligence platform (React/TypeScript, FastAPI) with electricity-bill analytics, regional benchmarking, and forecasting visualizations.
- Architected an agentic RAG pipeline with LangGraph and LangChain for query routing, evidence retrieval, and claim validation across cloud and self-hosted LLM providers.
- Quantified electricity price elasticity using DoWhy and EconML Double Machine Learning, controlling for weather factors (HDD/CDD) to isolate causal effects; delivered demand forecasts via XGBoost, Prophet, and SARIMAX.

- Developed, trained, and tested a computer vision pipeline for detecting utility/electric poles in street-level imagery using YOLOv8-OBB, achieving **mAP@50 = 0.95**, **Precision = 0.92**, **Recall = 0.90**.
- Designed an automated OBB annotation and QC workflow integrating foundation-model object discovery (DINO/SAM) to accelerate labeling and dataset generation.
- Implemented an active learning pipeline and tiled/sliced inference to prioritize low-confidence detections and preserve small pole features across mobile and Street View imagery.

Professional Service

Hackathon Judge — TRAVEL-WIZARD: Designing the Future of Conversational Travel

Conference Representative, NJLM Conference — Presented CESAC sustainability analytics research to policymakers and industry stakeholders.

PROJECTS

EfficientNetV2-Based Stage-Adaptive Attention Network for Microplastic Segmentation
2025 – October 2025

Deep Learning — Computer Vision — Environmental AI

- Designed a stage-adaptive attention encoder-decoder network using EfficientNetV2-B3 to address severe class imbalance and multi-scale variability in microscopy images.
- Integrated attention mechanisms, residual connections, and multi-scale feature fusion to improve detection of small and irregular microplastic particles.
- Applied hybrid loss functions (Dice + BCE / Focal Loss) and advanced preprocessing to improve robustness under extreme foreground sparsity and noise.
- Evaluated performance using Dice, IoU, Precision, and Recall (Dice: **0.7932**) on real-world microscopy datasets.

DisasterAI: Automated Damage Assessment & Report Generation System

Multimodal AI — LLM Systems — Computer Vision

- Developed a fully offline, privacy-preserving multimodal AI system for generating structured disaster damage reports from satellite imagery and structured inputs.
- Integrated CV pipelines (OpenCV + segmentation models) with a local LLM (Qwen via Ollama) and prompt-engineered pipelines to produce consistent, low-hallucination insurance-style reports.
- Built CLI and Streamlit interfaces and an evaluation pipeline measuring latency, severity extraction accuracy, and report completeness.
- Optimized system for CPU-only environments (4GB RAM), enabling scalable, low-cost deployment without cloud APIs.

Certifications

- Northeast Big Data Innovation Hub — Explorer Transportation Data Science Project
- NPTEL — Python for Data Science
- NPTEL — Database Management Systems
- Coursera — Data Analysis and Visualization with Power BI

Publications

- *Compound Multi-Hazard Assessment under CMIP6 Climate Scenarios: Tracking Seasonal Flood-Landslide and Drought-Fire Interactions Across Nepal*, **Natural Hazards**, Vol. 122, No. 14, Article 542 (2026).
- **2024 4th Asian Conference on Innovation in Technology (ASIANCON)**: *Enhancing Web Security Through Machine Learning: A Random Forest Approach to Malicious URL Detection*, pp. 1–6

- **Congress on Control, Robotics, and Mechatronics:** *Predicting Real Estate Price: A Comparative Analysis of Linear Regression and Support Vector Machine*, pp. 123–134
- Journal of Chemical Health Risks: *Ovarian Cancer Subtypes Reimagined: Leveraging CNNs for Advanced Classification and Outlier Detection*, Vol. 13, No. 6, Dec 2023
- *Speak Safe: An AI System for Robust Voice Authentication with Deepfake Detection and Noise Reduction*, ResearchSquare (preprint), 2026